

PROFILE

Data Scientist and Economic Researcher with a PhD in Economics, Management and **Statistics** and a Master's in **Big Data Analytics & Social Mining**, 5+ years delivering end-to-end data science for research projects ending in academic output and general and non-technical audience. Designs and ships predictive and machine learning models. Text analysis with NLP and LLM pipelines. Advanced Monte Carlo simulation, scenario and sensitivity analysis. End-to-end data engineering and interactive dashboard creation. Owns the full cycle of data management: ingestion from REST APIs and financial platforms, ETL and feature engineering, model training and evaluation. Dockerisation and cloud deployment, reporting and data storytelling for non-technical stakeholders. Domains: risk management, data science, public finance, judicial efficiency, media analytics, public health; solutions adopted by EU-funded institutions and cited in EU policy analysis. Italian native speaker, professional fluency in English.

EXPERIENCES

Researcher RTD (a) University of Palermo

Mar 2023 — Mar 2026

Project GRINS – Growing Resilient, Inclusive and Sustainable

- Improved and engineered Scenario - stochastic simulation engine for **Stochastic Debt Sustainability Analysis (SDSA)** from economists' fiscal questions, with **data ingestion from financial repositories (FactSet, Refinitiv Eikon)**; academic publication.
- Delivered and engineered as full-stack WhatDSApp, an interactive web app (Dash, Plotly, Docker) deployed to the AMELIA cloud platform and adopted by GRINS consortium partners for live **debt scenario analysis**, with a drag-and-drop multi-sheet **Excel ingestion pipeline**. App's analytics with **fan charts**, **efficient-frontier CVaR** plots, and **sensitivity analysis**.
- Produced academic **papers** and **periodic reporting** on **public finance and debt markets** of European Countries, and **budget analysis** of the US One Big Beautiful Bill Act for Bruegel.
- Research on network analysis, economic inequality, and analysis of performance of court houses. Use of **LLM** for **ESG scoring**.
- Tenure track in Mathematics for Business, calculus and financial mathematics.

Research Fellow University of Palermo

Apr 2022 — Mar 2023

Project Just Smart – Smart Justice

- Managed data retrieval from Italian judicial repositories (SICID) for western Sicily courthouses, and built a pipelines to measure the efficiency of judicial processes and deliver data-driven recommendations to court administration.
- Applied queueing analysis and survival models to judicial processes, quantifying bottlenecks and processing times that fed data-driven recommendations for court administration.

Research Fellow Ca' Foscari University

Apr 2021 — Apr 2022

Project H2020 Grant 101004488 – EUMEPLAT

- Designed data clustering on European media History to measure the harmonization of European media markets over 40 years.
- Built a public interactive dashboard (R Shiny, shinyapps.io) with cross-country choropleth, and descriptive statistics of the European media market (Press, Radio, TV, OTT, ICT).
- Lecturer for the module Advanced Application of Computer Science for Cultural Heritage, network analysis, and R programming.

PhD in Economics, Management and Statistics University of Messina, XXXIII Cycle (EQF 8)

Oct 2017 — Dec 2020

- Ran large-scale NLP on Italian Twitter data (sentiment analysis, text classification, REST API ingestion pipeline) to detect vaccination hesitancy and measure the impact of political communication on vaccination propensity; published in PLoS ONE 2021.
- Data Retrieval - streaming - from REST API (Tweepy) and data pipeline for massive social media data collection and analysis of the impact of covid-19 on the Italian populations response to pandemic.
- Thesis: "Disorientation towards routine immunization, COVID-19 anxiety, and propensity to vaccinate. Analysis based on Twitter data in Italy".
- Teaching in Business Statistics and Data Science for Demographic Processes, topic: R for data analytics, wrangling, and visualisation.

Internship RJC Soft s.r.l. - Pisa

Sep 2017 — Nov 2017

- Agent-based modelling and simulation of Pisa's urban traffic system using Python and QGIS.

ADDITIONAL TRAINING

Summer School in Network Econometrics, *Italian Society of Econometrics – Ca' Foscari University of Venice*

Jun 2022

Network extraction, finance application of multi-layer network models.

Fundamentals of Digital and Computational Demography, *Max Planck Institute for Demographic Research, Rostock*

Dec 2019

Tools to improve demographic and social science analysis, with a focus on interpreting new kinds of big data.

Summer School in Data Science and Epidemic Models, *University of Trento*

Jul 2019

Complex data analysis, digital tracking, human mobility, computational methods and social media analytics.

Summer School on Clustering, Data Analysis and Visualization of Complex Data, *University of Catania*

May 2018

Clustering, classification and visualisation of large datasets; numerical analysis and cloud-computing platforms.

EDUCATION

II Level Master's in Big Data Analytics & Social Mining , <i>University of Pisa</i> (EQF 7) Thesis: Traffic Analysis in Urban Environments via GPS Trajectories.	Jan 2017 — May 2018
Master's Degree in Economics , <i>University of Pisa, Scuola Superiore Sant'Anna</i> (EQF 7) Thesis: Option Pricing with Monte Carlo and Variance Reduction Technique.	Mar 2013 — Jul 2016
Bachelor's Degree in Economics and Finance , <i>University of Palermo</i> (EQF 6) Thesis: Loan Transfer and Securitisation.	Sep 2006 — Nov 2012

SKILLS

Programming & Development	Python (Pandas, NumPy, Scikit-learn, SciPy, Matplotlib), 5+ years in production, R (Shiny). SQL schema design, joins, analytical extraction. Spark MLlib for distributed ML. MATLAB, GAMS, Stata, Knime. Git/GitHub lifecycle: versioning, branching, code review. Linux administration, shell scripting, SSH, VS Code, Jupyter.
Machine Learning & Modelling	Supervised and unsupervised learning: regression, classification, clustering (k-means), feature engineering, PCA. Time-series forecasting, bootstrap CI, maximum likelihood. Ensemble methods (Random Forest, Gradient Boosting), optimisation, anomaly detection. Deep learning with Transformers, PyTorch, TensorFlow. Model validation and monitoring.
NLP, LLM & Generative AI	spaCy, NLTK, Gensim, HuggingFace Transformers, Word2Vec/TF-IDF for text classification, NER, topic recognition and sentiment analysis on Italian and multi-language corpora. Local deployment of open-source LLMs (Qwen, Deepseek, Gemma) via Ollama, Open WebUI, vLLM; supervised fine-tuning, prompt engineering, LLM integration into production pipelines. Claude Code (CLI) for agentic development.
Data Engineering, Big Data & Cloud	End-to-end ETL pipelines: ingestion, transformation, analysis, output. Relational, spatial and NoSQL databases (SQL/MySQL, PostgreSQL/PostGIS, MongoDB). Data retrieval from REST APIs, financial platforms, multi-sheet Excel. Docker containerisation, cloud deployment on AMELIA, async job orchestration. Data quality, governance. Hands-on experience with Apache Spark, Hadoop (MapReduce), Hive.
BI, Reporting & Visualisation	Production dashboards and advanced reporting: Plotly/Dash (WhatDSApp) and R Shiny (EUMEPLAT). Multi-panel and drill-down views, fan charts, choropleth visualisation. Short notes and in-depth reports, data storytelling for non-technical audiences. MS Office (Excel, PowerPoint), $\LaTeX$, Beamer, Bootstrap 4, HTML/CSS.
Quantitative & Risk Modelling	Monte Carlo simulation with variance reduction, scenario and sensitivity analysis, Stochastic Debt Sustainability Analysis, option pricing. Tail-risk optimisation (CVaR, efficient frontier), ESG risk factor modelling, public finance and debt-market analysis. Use of FactSet and Refinitiv Eikon. Econometric modeling with Stata and network analysis.

SELECTED PUBLICATIONS

- 2026** *Statistically Validated Lasso for Gaussian Graphical Models in Financial Networks* Numta 06/2026, pp35. numta.org
- 2025** *Are bad governments a threat to sovereign defaults? Effects of political risk on debt sustainability*. WP 01/2025. [Bruegel.org](https://bruegel.org)
- 2025** *How much of a threat to US debt sustainability is Trump's One Big Beautiful Bill Act?* Analysis Report, Bruegel. [Bruegel.org](https://bruegel.org)
- 2025** *ESG factors and sovereign debt*. *Economia Italiana* 46(3), 145–164. economaiitaliana.org
- 2024** *Debt Sustainability in the Context of Population Ageing: A Risk Management Approach*. *Risks* 12(12). mdpi.com
- 2022** *H2020 European Media Platform – EUMEPLAT – Data Clustering Reports*. eumeplat.eu
- 2021** *Evidence of disorientation towards immunization on online social media*. *PLoS ONE*. journals.plos.org

SOFTWARE AND TOOLS RELEASED

WhatDSApp: full-stack decision-support app for sovereign debt scenario analysis. Stochastic Debt Analysis models for financing optimisation, fan charts and efficient frontier, plug-and-play ingestion, data from Refinitiv Eikon and FactSet; deployed on the AMELIA cloud platform, adopted by GRINS consortium partners. ameliadp.grins.it

EUMEPLAT Dashboard: interactive dashboard for the EUMEPLAT project, data clustering reports. eumeplat.shinyapps.io

HOBBIES

Cycling, running, bricolage, reading, programming for fun, car simulations.